

CSC6033 – Week 5

In-Class Exercise E#5

CFG and PDA for Palindromes (Even and Odd Length)

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Alphabet: $\Sigma = \{a, b\}$ **Slides referenced:** 8, 9, 14, 15, 28, 29

1. Introduction

This document merges the *even-letters-palindrome* (Slides 8, 14, 28) and *odd-letters-palindrome* (Slides 9, 15, 29) examples into a single solution that accepts *all* palindromes over $\{a, b\}$, i.e.,

$$L = \{ w \in \{a, b\}^* \mid w = w^R \},$$

covering both even and odd lengths. We present (i) a compact CFG and (ii) a single nondeterministic PDA that recognizes the same language.

2. Context-Free Grammar (CFG)

2.1 Grammar $G = (V, \Sigma, R, S)$

- Terminals: $\Sigma = \{a, b\}$
- Nonterminals: $V = \{S\}$
- Start symbol: S
- Productions R :

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \varepsilon$$

2.2 Rationale

- aSa and bSb enforce mirrored endpoints, building palindromes outward-in.
- a and b permit a single unmatched middle symbol (odd length).
- ε permits the empty word (even length).

2.3 Sample derivations

$$S \Rightarrow aSa \Rightarrow aaSaa \Rightarrow aabSbaa \Rightarrow aab\varepsilonbaa = \underline{aabbaa}$$

$$S \Rightarrow bSb \Rightarrow bab$$

$$S \Rightarrow \varepsilon$$

3. Pushdown Automaton (PDA)

We give an NPDA that pushes the first half of the input onto the stack, then nondeterministically *guesses* the midpoint. It either (i) switches to comparison without consuming a letter (even length), or (ii) consumes exactly one middle symbol without stack action (odd length). It then pops to match the mirrored second half. Acceptance occurs when the input is finished and only the bottom marker remains.

3.1 Formal definition

The PDA is $M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$ with:

- $Q = \{I, J, K, F\}$ (Initial, Push-mode, Compare-mode, Final)
- $\Sigma = \{a, b\}$
- $\Gamma = \{a, b, z\}$ (stack alphabet; z is the bottom marker)
- $q_0 = I$
- $F = \{F\}$

Transition function δ (complete list).

Init: $\delta(I, \varepsilon, \varepsilon) \ni (J, z)$

push bottom marker z and

Push mode: $\delta(J, a, z) = (J, az), \quad \delta(J, a, a) = (J, aa), \quad \delta(J, a, b) = (J, ab)$
 $\delta(J, b, z) = (J, bz), \quad \delta(J, b, a) = (J, ba), \quad \delta(J, b, b) = (J, bb)$

Midpoint (even): $\delta(J, \varepsilon, z) = (K, z), \quad \delta(J, \varepsilon, a) = (K, a), \quad \delta(J, \varepsilon, b) = (K, b)$

Midpoint (odd): $\delta(J, a, z) \ni (K, z), \quad \delta(J, a, a) \ni (K, a), \quad \delta(J, a, b) \ni (K, b)$
 $\delta(J, b, z) \ni (K, z), \quad \delta(J, b, a) \ni (K, a), \quad \delta(J, b, b) \ni (K, b)$

Compare / pop: $\delta(K, a, a) = (K, \varepsilon), \quad \delta(K, b, b) = (K, \varepsilon)$

Accept: $\delta(K, \varepsilon, z) = (F, \varepsilon)$

Notes.

- The machine is nondeterministic due to the two *midpoint* choices.
- ε -moves follow the style used in the lecture slides.
- Acceptance is by final state F after popping the bottom marker z .

3.2 TikZ diagram

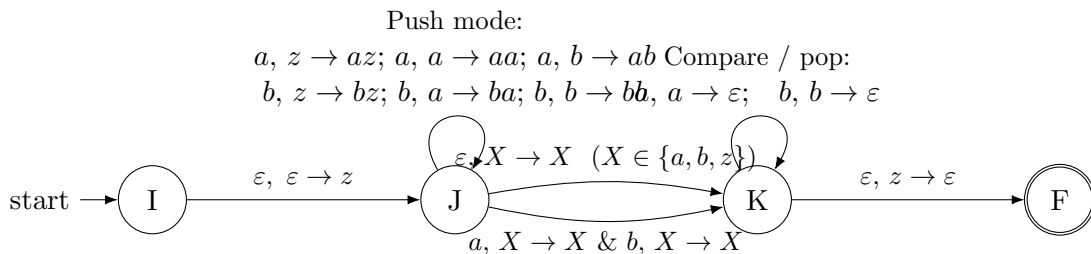


Figure 1: NPDA for all palindromes over $\{a, b\}$ (even and odd lengths).

3.3 Example runs (stack top on the right)

$w = abba$ (even).

1. Init: push z .
2. Push mode reads a , b and pushes; nondet. choose even midpoint: ε -switch to K .
3. Compare mode pops on b then a ; input consumed, stack has z .
4. $\varepsilon, z \rightarrow \varepsilon$ to F (accept).

$w = aba$ (odd).

1. Init: push z ; push a .
2. Take odd midpoint transition: consume middle b without stack change, switch to K .
3. Compare: read a , pop a ; input consumed, stack has z .
4. $\varepsilon, z \rightarrow \varepsilon$ to F (accept).

$w = abab$ (reject).

- Any path fails to match mirrored pops (a mismatch eventually occurs on top of the stack), so no accepting computation exists.

4. Correctness (brief sketch)

Soundness. Any accepted string w must consume symbols symmetrically in K , because a pop is only possible when the input symbol equals the stack top. As the stack stores the first half in order, the second half must be its reverse, hence $w = w^R$.

Completeness. For any palindrome w , choose the nondeterministic switch exactly at the midpoint—using ε for even length, or consuming the middle letter for odd length. Then each remaining input symbol equals the corresponding stack top, ensuring a successful sequence of pops until z and acceptance.

5. Remarks on Adaptation

- **CFG merge.** Even-palindrome grammar ($S \rightarrow aSa \mid bSb \mid \varepsilon$) and odd-palindrome grammar ($S \rightarrow aSa \mid bSb \mid a \mid b$) are unified as $S \rightarrow aSa \mid bSb \mid a \mid b \mid \varepsilon$.
- **PDA merge.** Start from the even-palindrome PDA and add transitions that *optionally* consume exactly one input symbol with no stack action before switching to compare-mode, thereby covering odd lengths too.
- **Non-determinism.** Both even and odd cases are handled via a nondeterministic guess of the midpoint (consistent with class slides).

6. Quick acceptance tests

- Accepted (even): ε , aa , bb , $abba$, $baab$, $bbaabb$
- Accepted (odd): a , b , aba , bab , $aabaa$, bbb
- Rejected: ab , abb , aab , $abab$